

On the Capacity of Interference Channels with Partial Codebook Knowledge

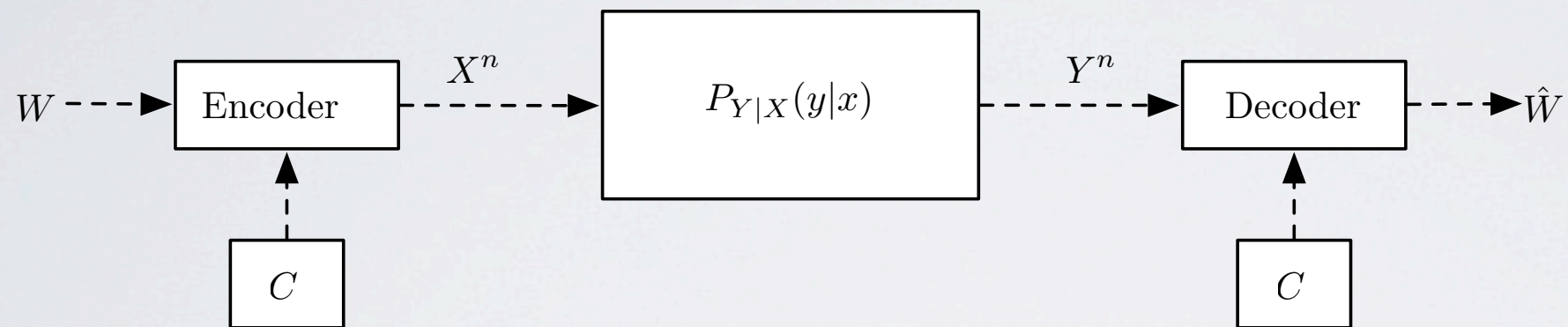
Alex R Dytso, Natasha Devroye, Daniela Tuninetti

The work of the authors was partially funded by NSF under award 1017436



CLASSICAL ENCODING

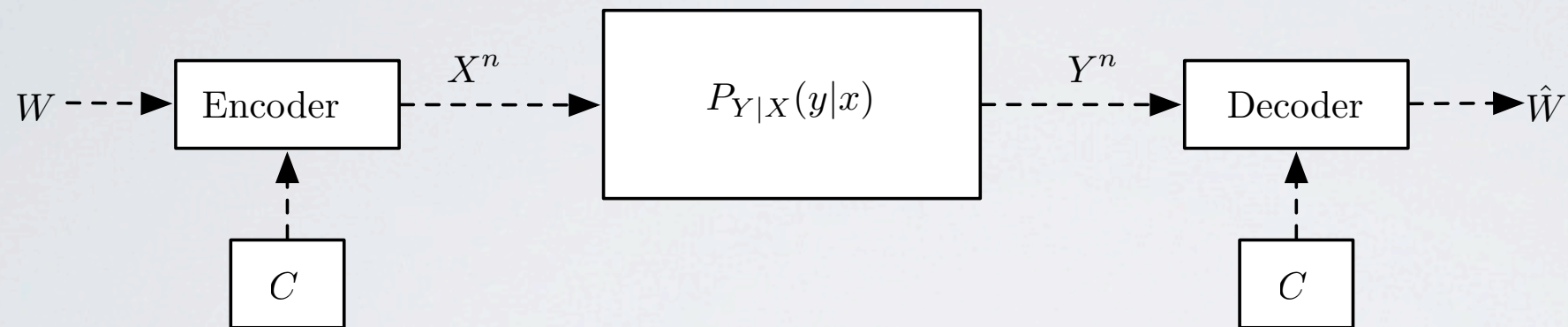
Point-to-Point



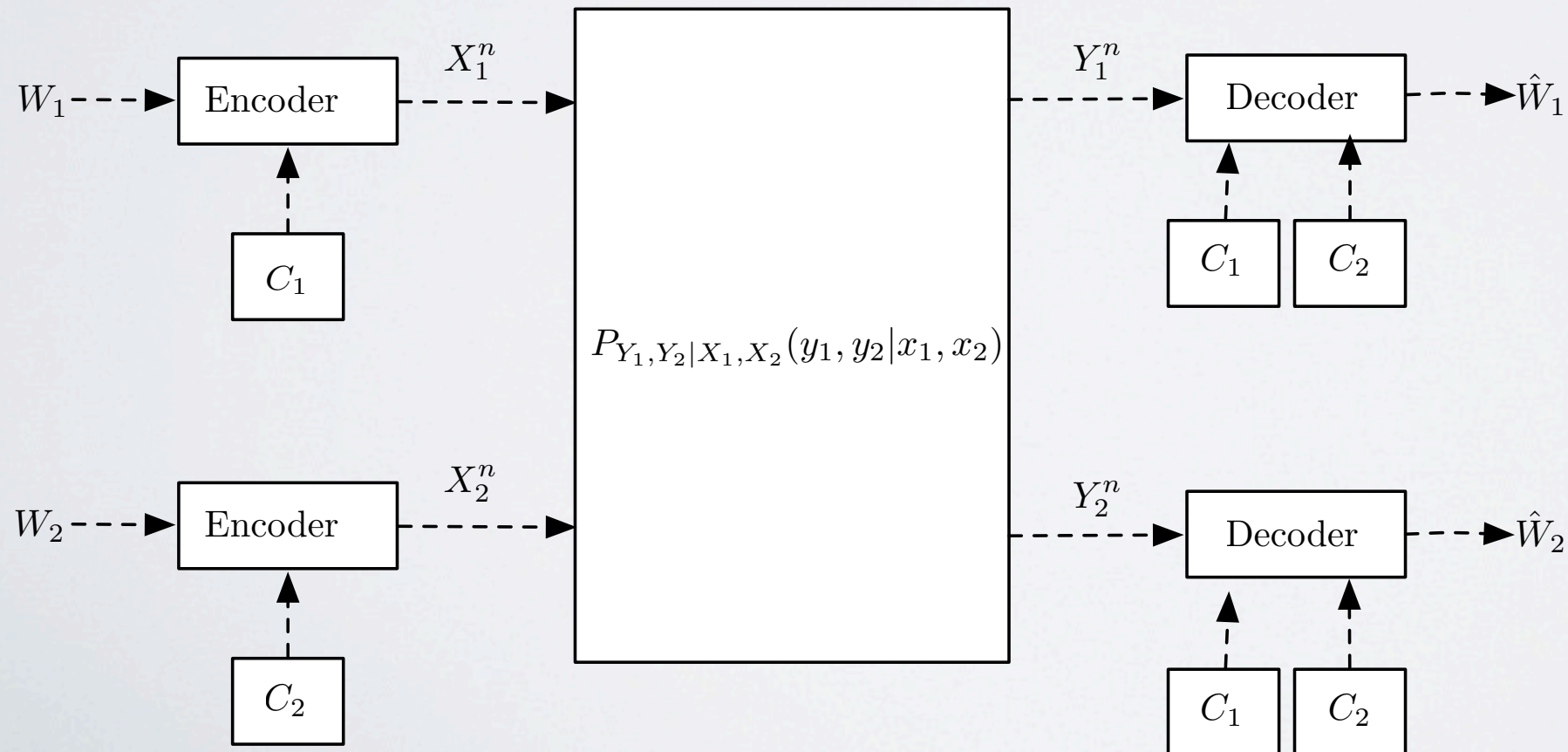
$$C = \begin{pmatrix} X^n(1) \\ X^n(2) \\ X^n(3) \\ \vdots \\ X^n(2^n R) \end{pmatrix}$$

CLASSICAL ENCODING

Point-to-Point

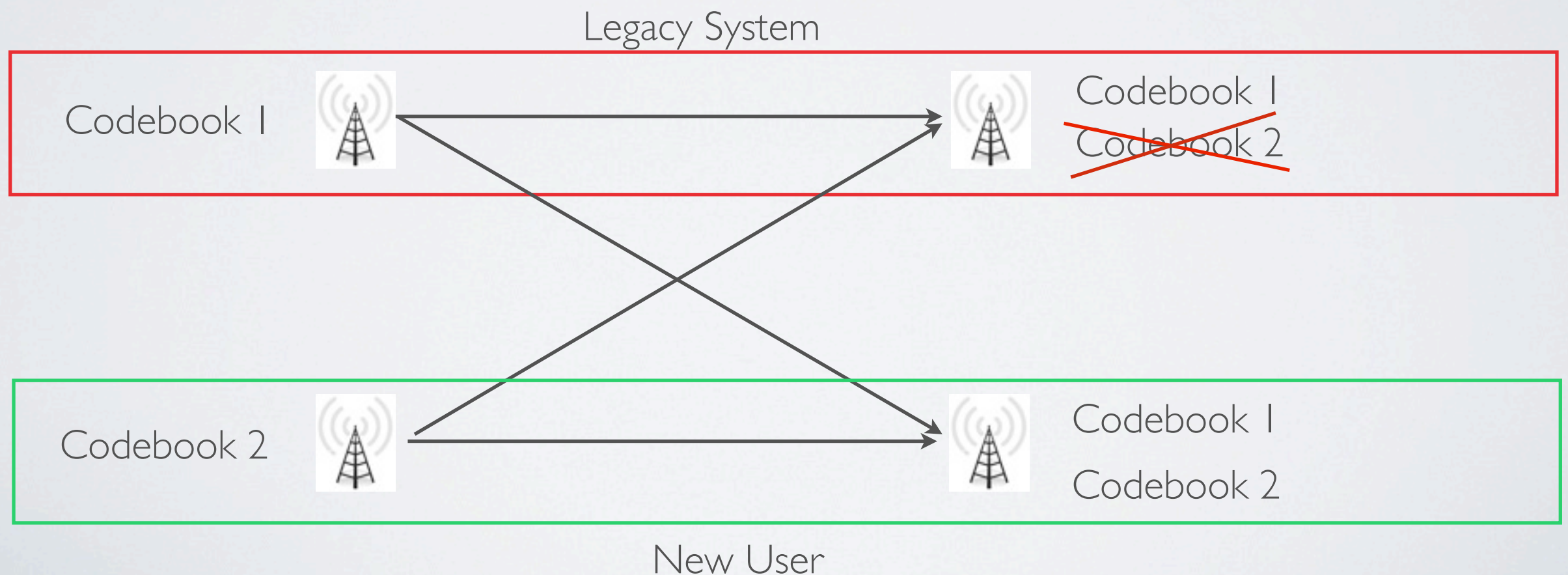


2-User



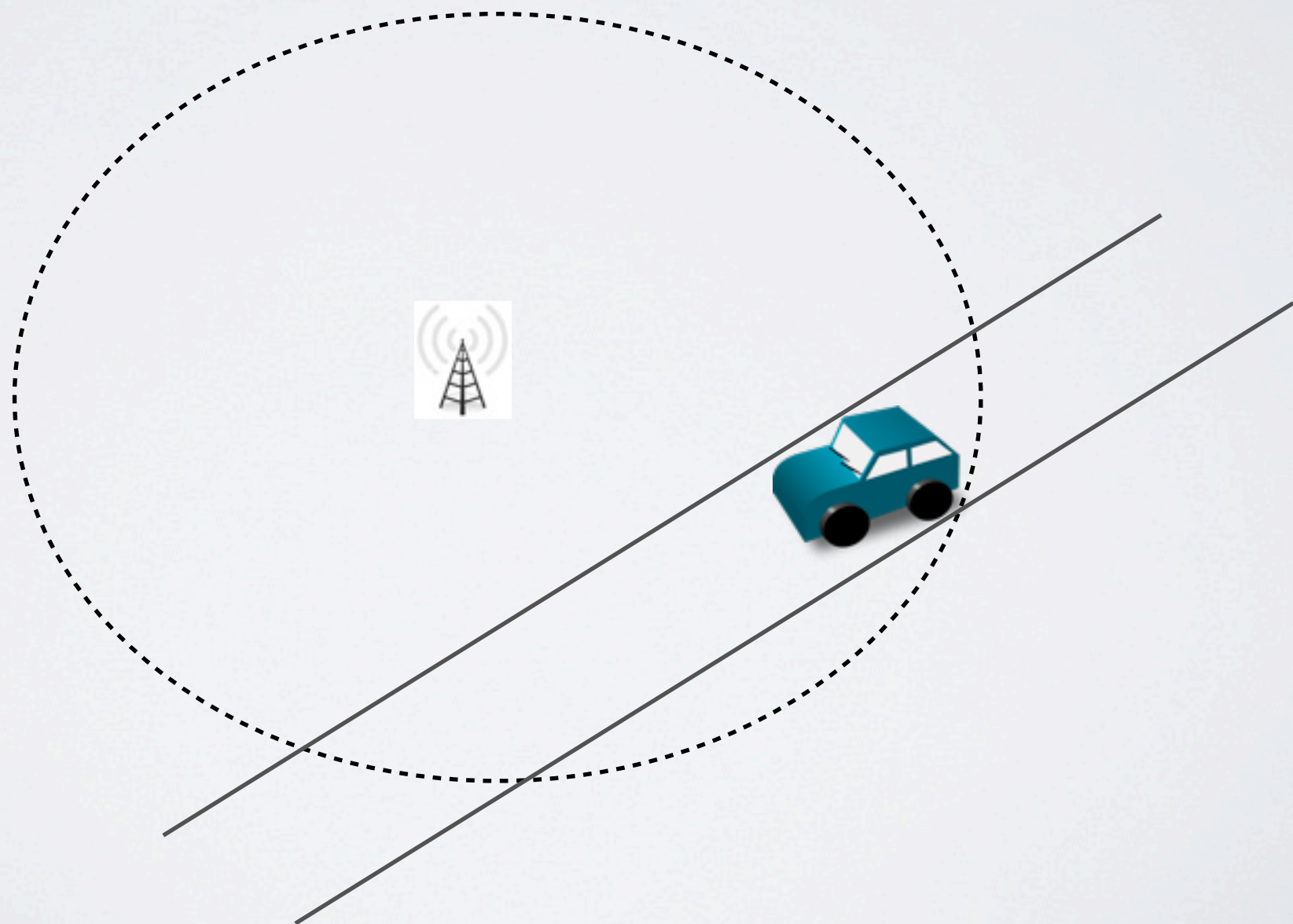
WHY THIS MIGHT BE UNPRACTICAL?

Legacy systems can NOT learn codebooks



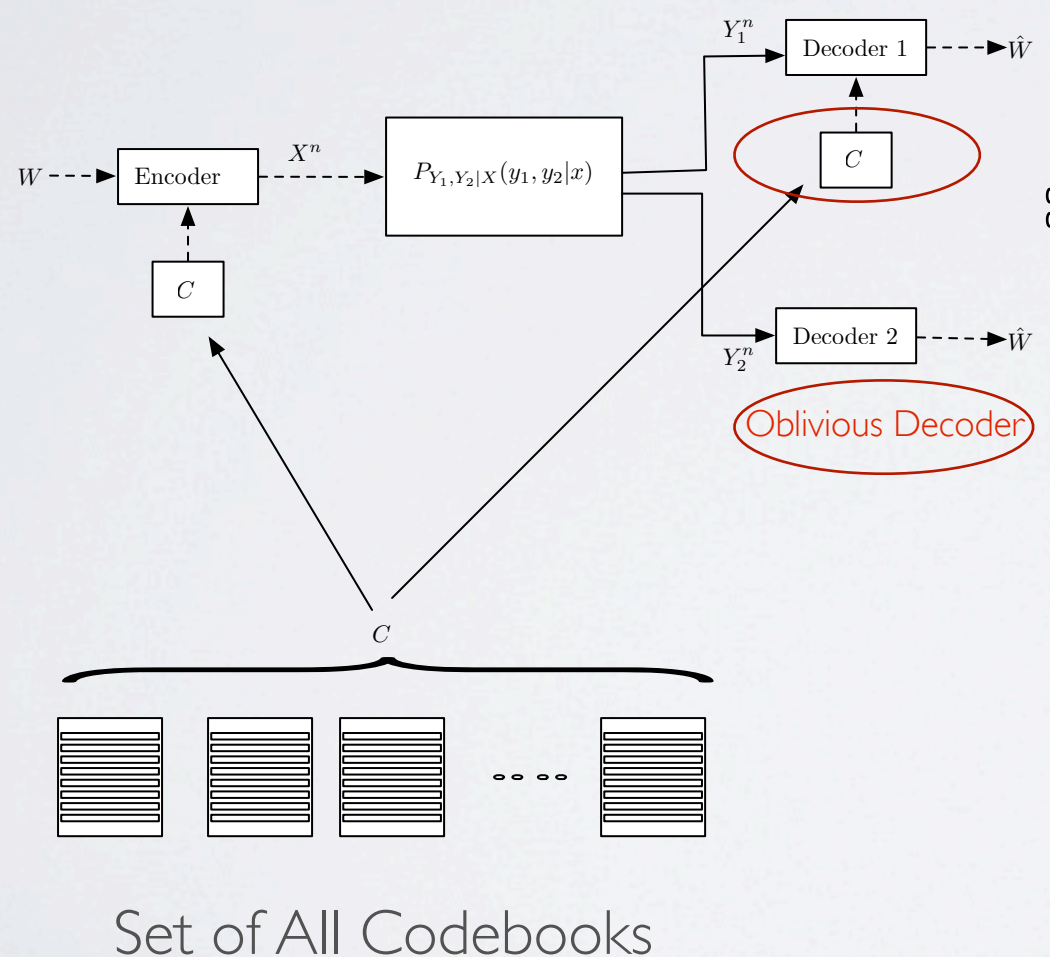
WHY THIS MIGHT BE UNPRACTICAL?

Mobility and Time Constrains



MODELING LACK OF CODEBOOK KNOWLEDGE

Random Encoding

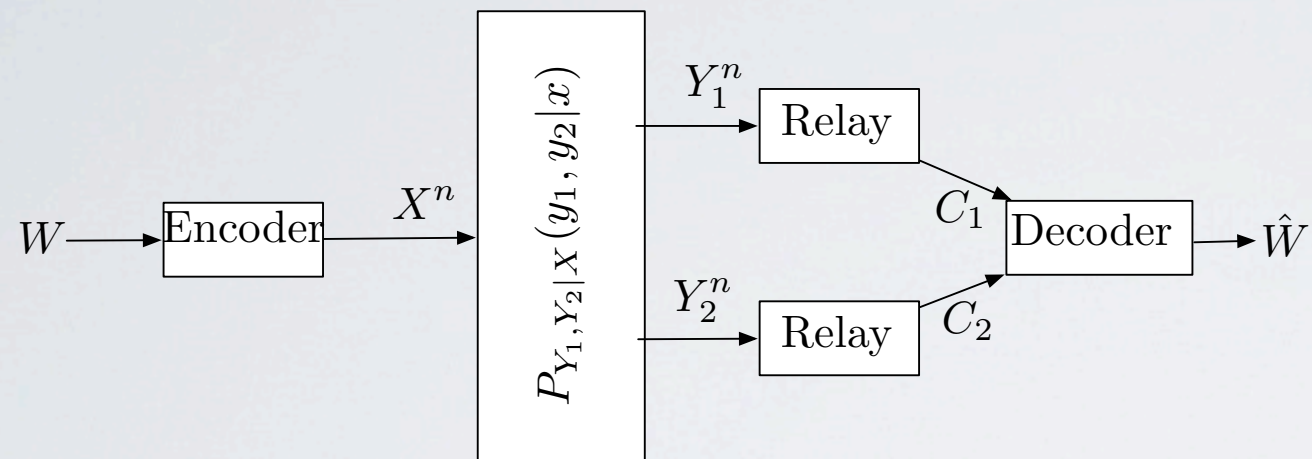


$$P_{X^n}(x^n) = \prod_{i=1}^n P_X(x)$$

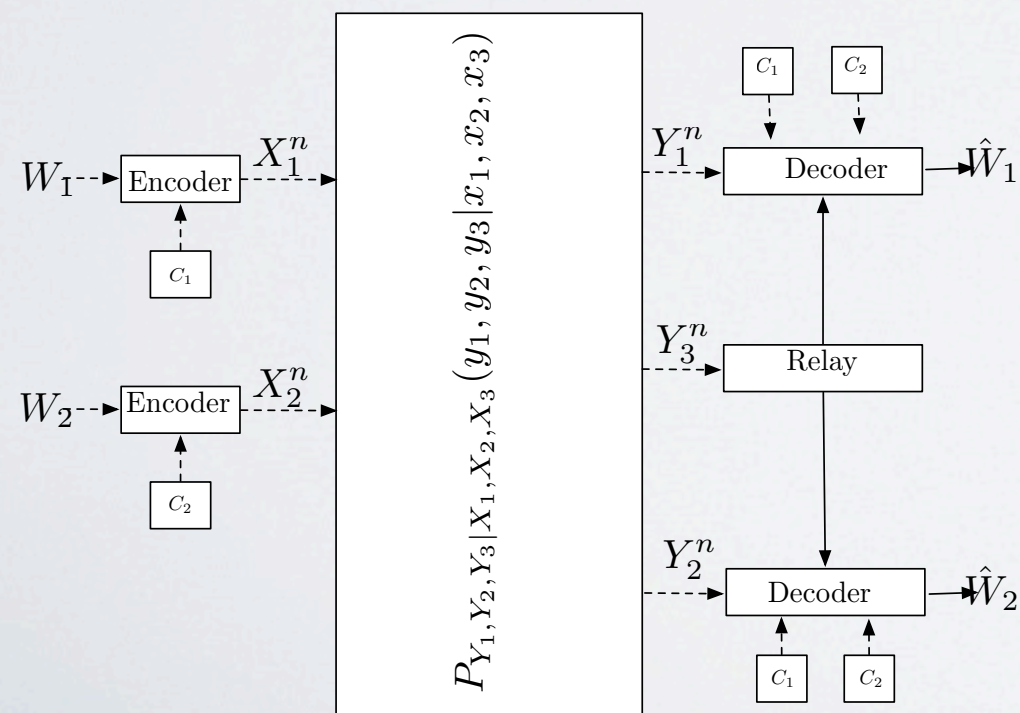
$$P_{X^n|C}(x^n|c) \quad \text{Not Product}$$

A. Sanderovich, S. Shamai, Y. Steinberg, and G. Kramer, "Communication via decentralized processing," IEEE Trans. Inf. Theory, vol. 54, no. 7, pp. 3008–3023, Jul. 2008.

RELATED WORK



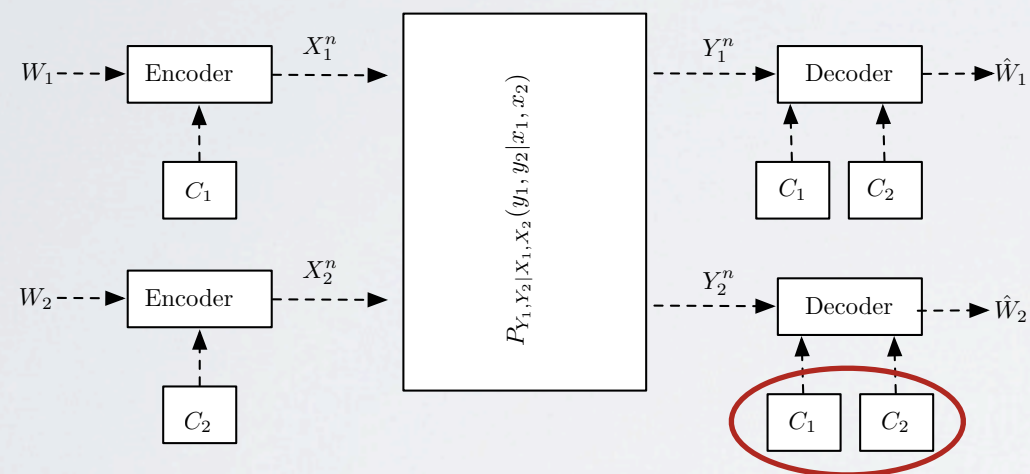
A. Sanderovich, S. Shamai, Y. Steinberg, and G. Kramer, "Communication via decentralized processing," IEEE Trans. Inf. Theory, vol. 54, no. 7, pp. 3008–3023, Jul. 2008.



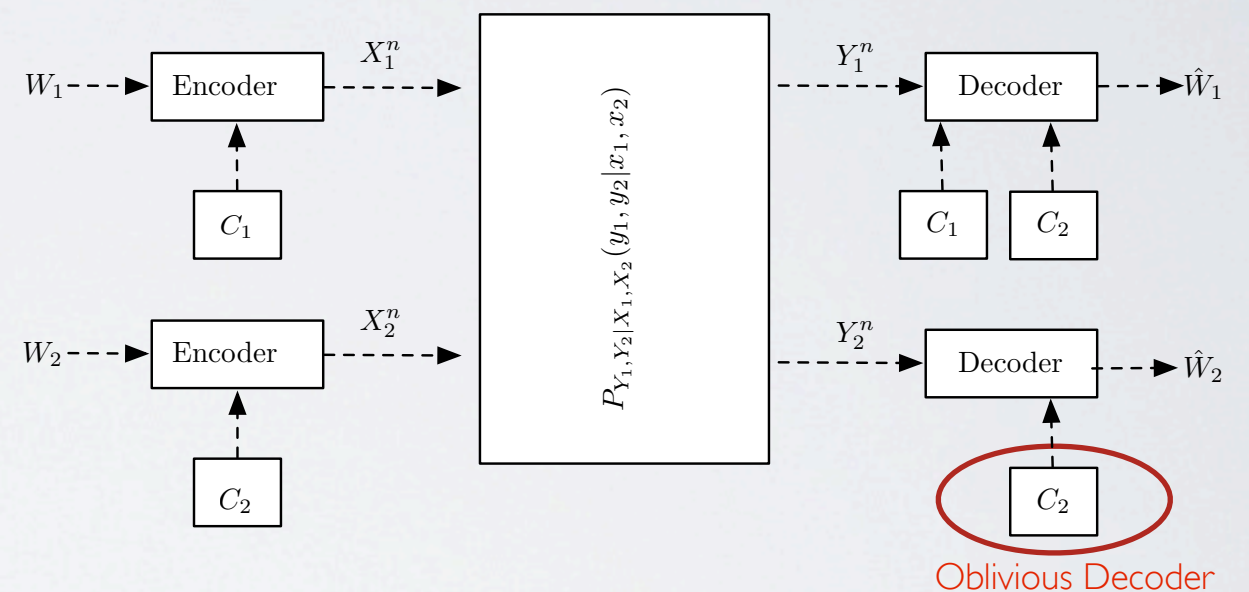
O. Simeone, E. Erkip, and S. Shamai, "On codebook information for interference relay channels with out-of-band relaying," IEEE Trans. Inf. Theory, vol. 57, no. 5, pp. 2880–2888, may 2011.

MODEL UNDER CONSIDERATION

Classical Interference Channel



Interference Channel with Partial Codebook knowledge

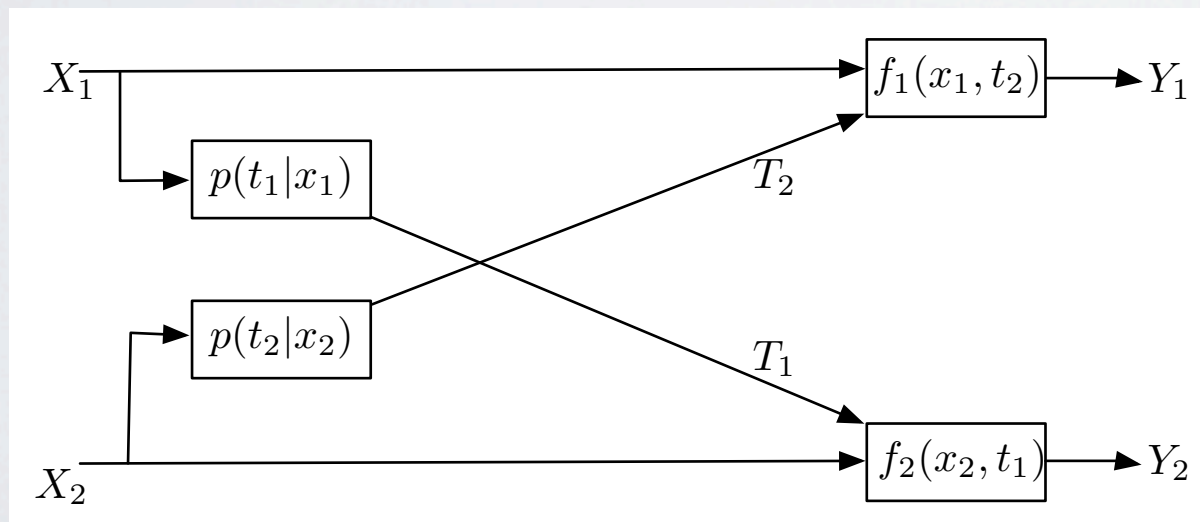


MODEL UNDER CONSIDERATION

We consider semi-deterministic injective interference channel

$$P_{Y_1 Y_2 | X_1 X_2}(y_1, y_2 | x_1, x_2) = \sum_{t_1, t_2} P_{T_1 | X_1}(t_1 | x_1) P_{T_2 | X_2}(t_2 | x_2) \delta(y_1 - f_1(x_1, t_2)) \delta(y_2 - f_2(x_2, t_1))$$

$$H(Y_1 | X_1) = H(T_2 | X_1) \text{ and } H(Y_2 | X_2) = H(T_1 | X_2)$$



E. Telatar and D. Tse, "Bounds on the capacity region of a class of interference channels," in Proc. IEEE Int. Symp. Inf. Theory, June 2007, pp. 2871–2874.

STRUCTURE OF INTERFERENCE

Lemma. *The output of the oblivious decoder has a product distribution conditioned on the parameters of the known codebook, that is*

$$P_{Y_2^n | X_2^n C_2 Q^n}(y_2^n | x_2^n, c_2, q^n) = \prod_{i \in [1:n]} P_{Y_2 | X_2, Q}(y_{2i} | x_{2i}, q_i).$$

i.e. interference has no structure

This result is used in a key step of our converse

CONVERSE

Theorem (Region Ro). Any achievable rate pair must satisfy

$$R_1 \leq I(Y_1; X_1 | X_2, Q)$$

$$R_2 \leq I(Y_2; X_2 | Q)$$

$$R_1 + R_2 \leq H(Y_1 | Q) + H(Y_2 | U_2, Q) \\ - H(T_2 | X_2, Q) - H(T_1 | Q)$$

for some input distribution $P_{Q, X_1, X_2, U_2} = P_Q P_{X_1 | Q} P_{X_2 | Q} P_{U_2 | X_2}$

where U_2 is conditionally independent copy of T_2 given X_2

$$P_{U_2, T_2 | X_2, Q}(u_2, t_2 | x_2, q) = P_{T_2 | X_2, Q}(t_2 | x_2, q) P_{T_2 | X_2, Q}(u_2 | x_2, q)$$

and where $|Q| \leq 2$

CONVERSE

Theorem (Region Ro). Any achievable rate pair must satisfy

$$R_1 \leq I(Y_1; X_1 | X_2, Q) \quad \leftarrow \text{Cut Set Bound}$$

$$R_2 \leq I(Y_2; X_2 | Q)$$

$$R_1 + R_2 \leq H(Y_1 | Q) + H(Y_2 | U_2, Q) \\ - H(T_2 | X_2, Q) - H(T_1 | Q)$$

for some input distribution $P_{Q, X_1, X_2, U_2} = P_Q P_{X_1 | Q} P_{X_2 | Q} P_{U_2 | X_2}$

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$$P_{U_2, T_2 | X_2, Q}(u_2, t_2 | x_2, q) = P_{T_2 | X_2, Q}(t_2 | x_2, q) P_{T_2 | X_2, Q}(u_2 | x_2, q)$$

and where $|Q| \leq 2$

Key Steps:

$$nR_2 \stackrel{(a)}{\leq} I(W_2; Y_2^n | C_2) + n\epsilon_n$$

$$\stackrel{(b)}{\leq} I(X_2^n; Y_2^n | C_2) + n\epsilon_n$$

$$\stackrel{(c)}{=} \sum_{i \in [1:n]} I(X_{2i}; Y_{2i}) + n\epsilon_n$$

CONVERSE

Theorem (Region Ro). Any achievable rate pair must satisfy

$$R_1 \leq I(Y_1; X_1 | X_2, Q) \quad \leftarrow \text{Cut Set Bound}$$

$$R_2 \leq I(Y_2; X_2 | Q)$$

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for some input distribution $P_{Q, X_1, X_2, U_2} = P_Q P_{X_1 | Q} P_{X_2 | Q} P_{U_2 | X_2}$

where U_2 is conditionally independent copy of T_2 given X_2

$$P_{U_2, T_2 | X_2, Q}(u_2, t_2 | x_2, q) = P_{T_2 | X_2, Q}(t_2 | x_2, q) P_{U_2 | X_2, Q}(u_2 | x_2, q)$$

and where $|Q| \leq 2$

Key Steps:

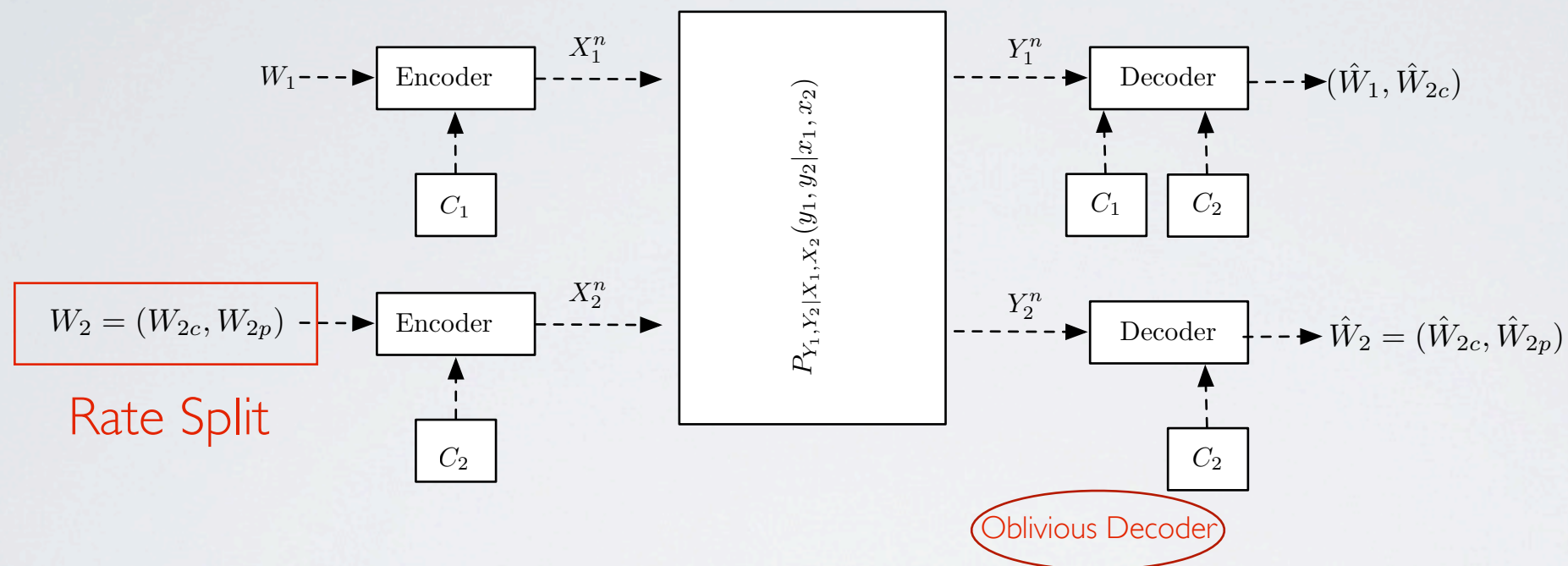
$$nR_2 \stackrel{(a)}{\leq} I(W_2; Y_2^n | C_2) + n\epsilon_n$$

$$\stackrel{(b)}{\leq} I(X_2^n; Y_2^n | C_2) + n\epsilon_n$$

$$\stackrel{(c)}{=} \sum_{i \in [1:n]} I(X_{2i}; Y_{2i}) + n\epsilon_n$$

Sum Rate requires injective assumption

ACHIEVABLE STRATEGY



Theorem (Region Ri). The following rate region is achievable

$$R_1 \leq I(X_1; Y_1 | U_2, Q)$$

$$R_2 \leq I(X_2; Y_2 | Q)$$

$$R_1 + R_2 \leq I(X_1, U_2; Y_1 | Q) + I(X_2; Y_2 | U_2, Q)$$

for all distributions in $P_{Q, X_1, X_2, U_2} = P_Q P_{X_1 | Q} P_{X_2 | Q} P_{U_2 | X_2}$

} Special Case of HK with rate split at user 2 ONLY

CONSTANT GAP CAPACITY

Theorem. If $(R_1, R_2) \in \mathcal{R}_O$, then

$$(R_1, R_2) \in \mathcal{R}_O \implies (R_1 - \overset{\text{Gap}}{I(X_2; T_2 | U_2, Q)}, R_2) \in \mathcal{R}_i.$$

Real valued Gaussian IC

$$\begin{aligned} Y_1 &= h_{11}X_1 + h_{12}X_2 + Z_1 & \mathbb{E}[X_i^2] &\leq 1 & Z_i &\sim \mathcal{N}(0, 1) \\ Y_2 &= h_{21}X_1 + h_{22}X_2 + Z_2 \end{aligned}$$

Theorem. If $(R_1, R_2) \in \mathcal{R}_O$, then $(R_1 - \overset{\text{Gap}}{1/2}, R_2)$ is achievable.

LINEAR DETERMINISTIC MODEL

A. Avestimehr, S. Diggavi, and D. Tse, "A deterministic approach to wireless relay networks," Proc. Allerton Conf. Commun., Control and Comp., Sep. 2007.

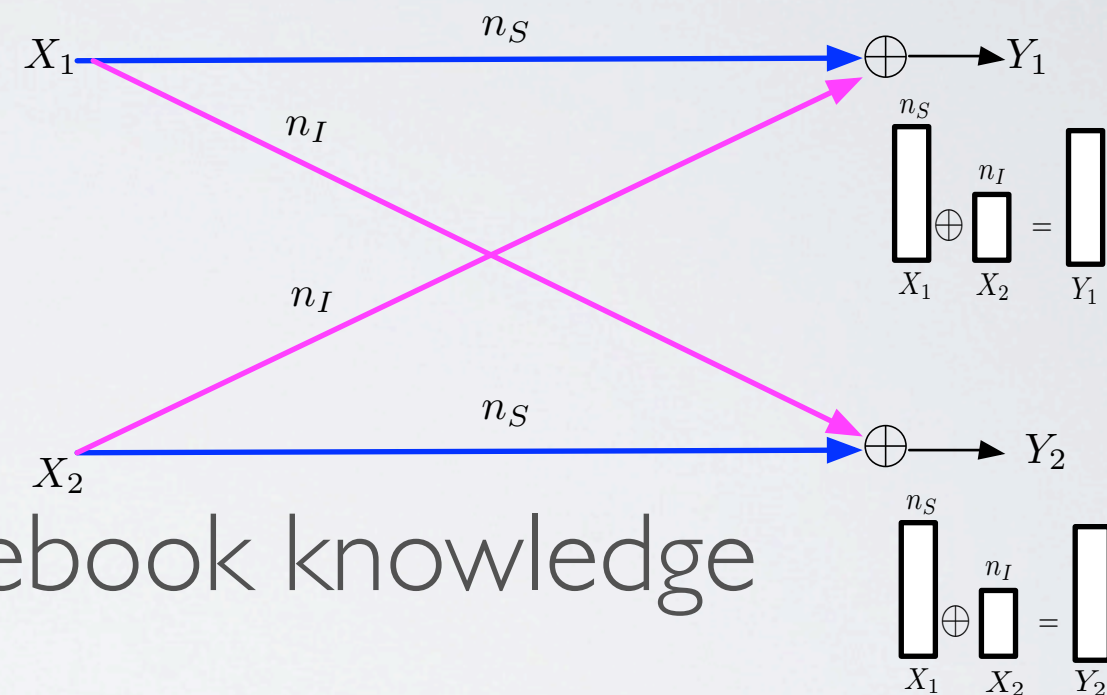
Model

$$Y_1 = \mathbf{S}^{q-n_S} X_1 + \mathbf{S}^{q-n_I} X_2$$

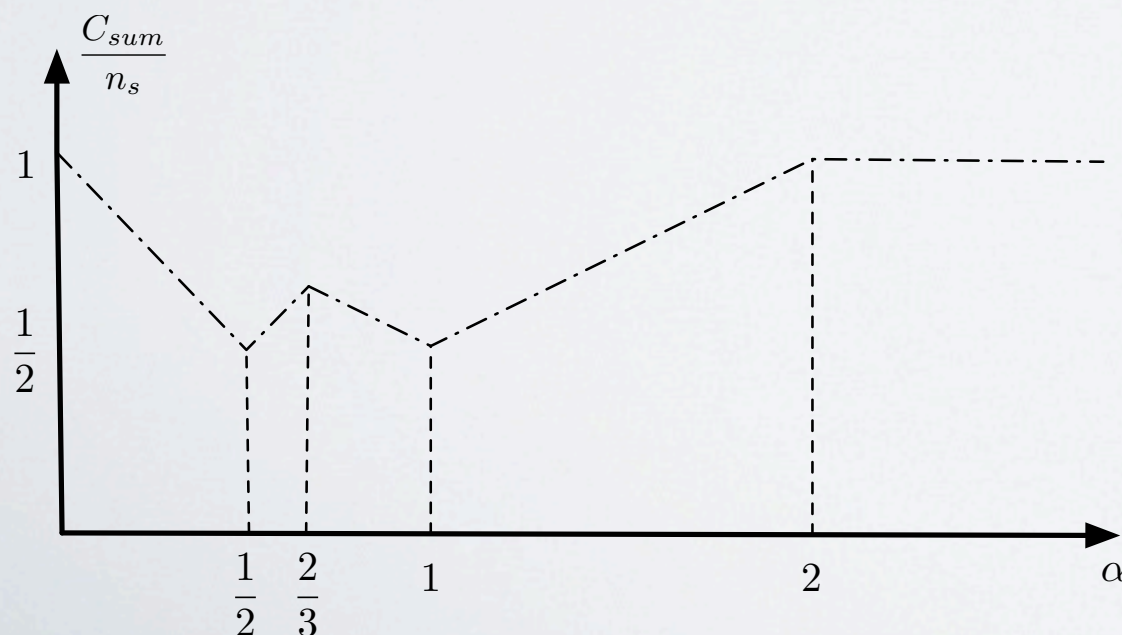
$$Y_2 = \mathbf{S}^{q-n_I} X_1 + \mathbf{S}^{q-n_S} X_2$$

$$\alpha = \frac{n_I}{n_S} \text{ Interference level}$$

Approximation to Gaussian



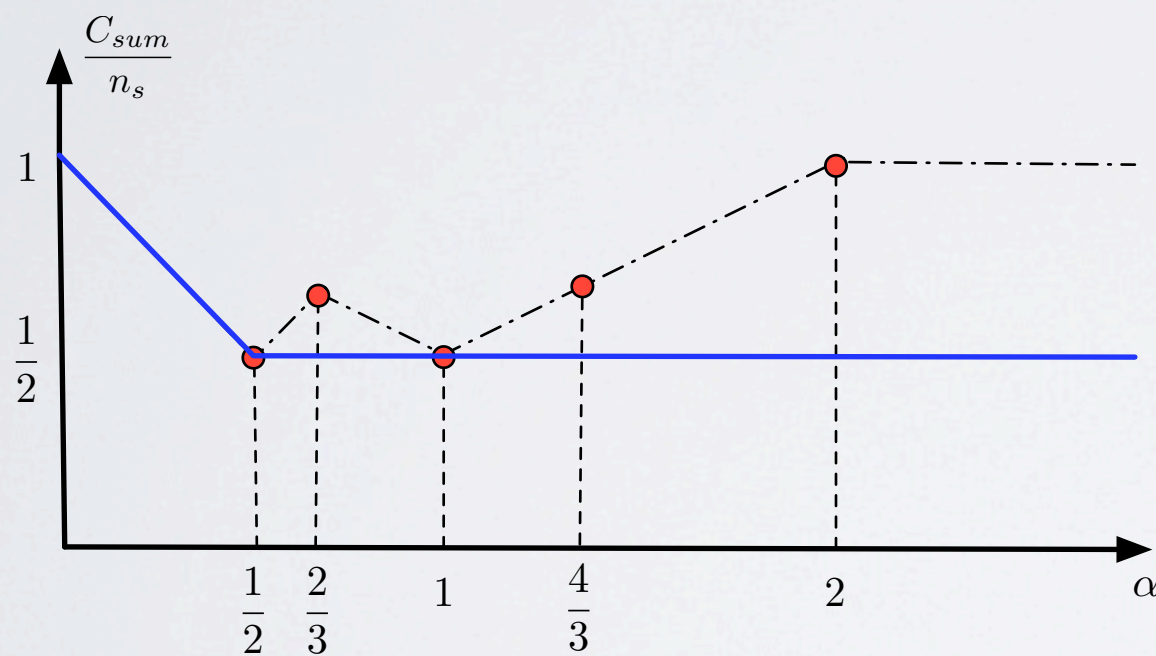
Sum Capacity with Full codebook knowledge



Maximizing distribution is i.i.d. Bernoulli(1/2)

G. Bresler and D. Tse, "The two-user gaussian interference channel: A deterministic view," European Transactions in Telecommunications, vol. 19, pp. 333–354, Apr. 2008.

LINEAR DETERMINISTIC MODEL WITH PARTIAL CODEBOOK KNOWLEDGE



α	Probability mass function
$\frac{1}{2}$	$P_{X_1} = [0.5, 0, 0.5, 0]$ $P_{X_2} = [0, 0.5, 0, 0.5]$
$\frac{2}{3}$	$P_{X_1} = [0, 0, 0.25, 0.25, 0, 0, 0, 0.25, 0.25]$ $P_{X_2} = [0, 0, 0.25, 0.25, 0, 0, 0, 0.25, 0.25]$
1	$P_{X_1} = [0, 0, 0.5, 0.5]$ $P_{X_2} = [0, 0.5, 0, 0.5]$
$\frac{4}{3}$	$P_{X_1} = [0, 0, 0, 0, 0, 0.25, 0, 0.25, 0, 0, 0, 0, 0.25, 0, 0.25]$ $P_{X_2} = [0, 0, 0, 0.25, 0, 0.25, 0, 0, 0, 0, 0, 0, 0.25, 0, 0.25]$
2	$P_{X_1} = [0, 0.5, 0, 0.5]$ $P_{X_2} = [0, 0.5, 0, 0.5]$

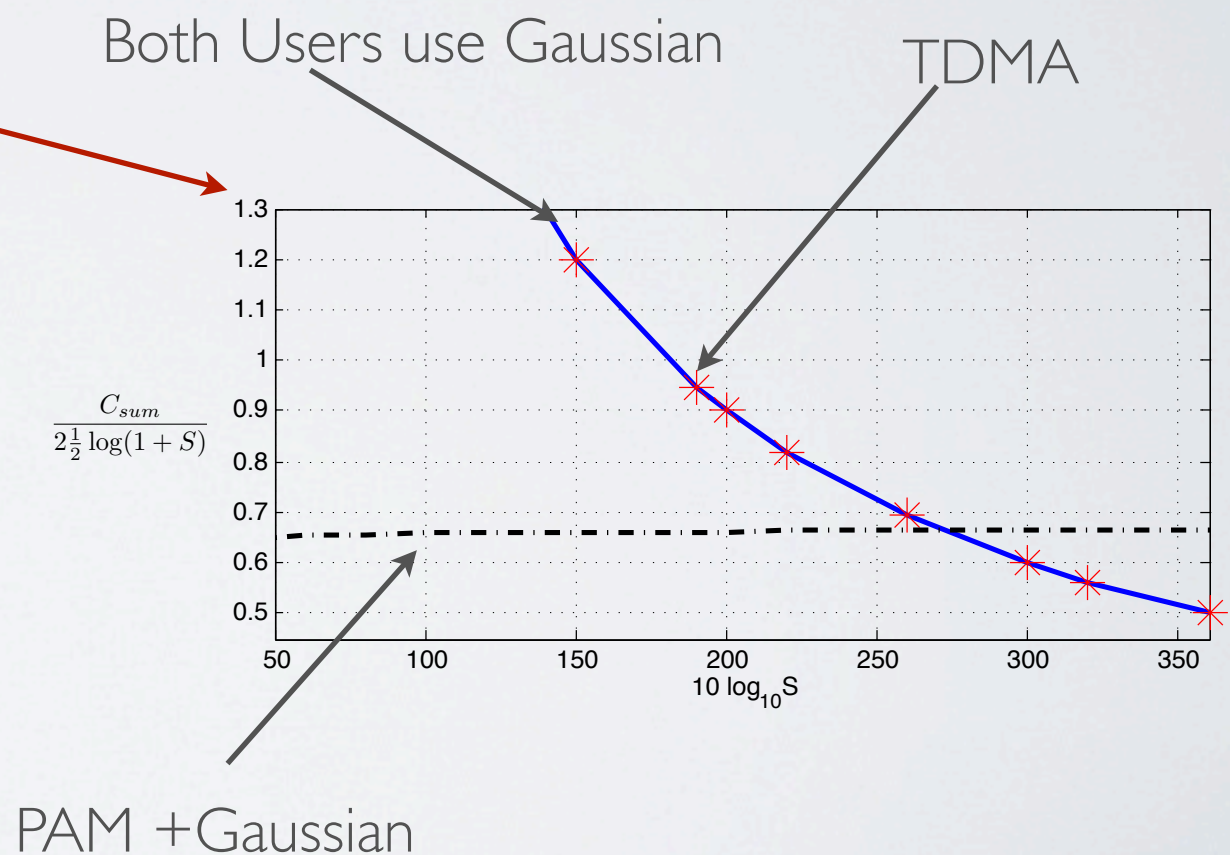
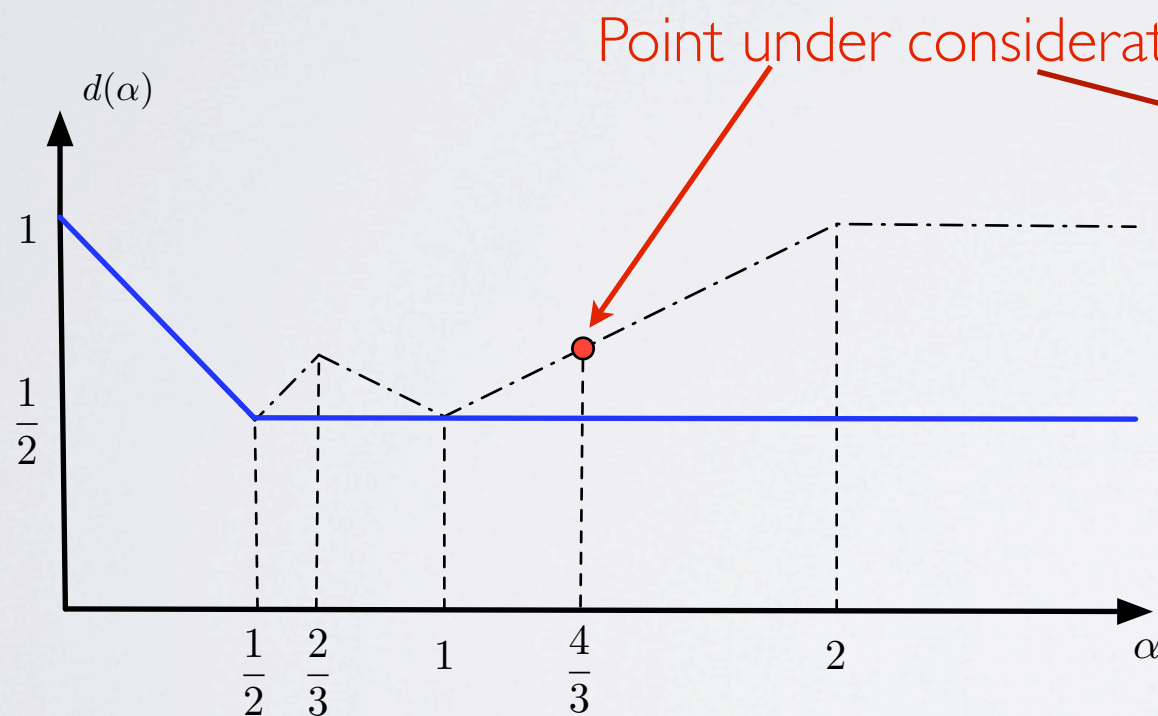
Maximizing distribution is NOT i.i.d. Bernoulli(1/2)

GENERALIZED DEGREES OF FREEDOM

gDof

$$g(\alpha) := \lim_{S \rightarrow +\infty} \frac{C_{sum}}{2^{\frac{1}{2}} \log(1+S)}$$

$$\alpha = \frac{\log I}{\log S}$$



CONCLUSION

- Introduce a new IC mode
- Show capacity to within a constant gap
- Show that in the case symmetric LD-IC partial codebook knowledge does NOT reduce sum-capacity
- Show that some parts of the W-curve of Gaussian IC can be achieved with partial codebook knowledge

THANK YOU